

Periodic Reassembly Inside a Quantitative Multiplicative Corridor: Exact Quantifiers, Zero Residue-Census Loss, and Non-Prefix Scope

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July 2026

Abstract

The quantitative two-point theorem of Tao and Teräväinen is uniform in the congruence modulus, residue and two shifts after its multiplicative functions and ambient scale have been fixed. We source-lock these quantifiers and derive a periodic reassembly principle that will be used on determinant-two Möbius fibers. If one progression of modulus Q carries a bounded multiplier of period R , and QR lies in the theorem's small-polylogarithmic modulus envelope, then resolving the multiplier into R residue classes creates no factor R . Each refined class has natural mass $N/(QR)$, so its R -fold census exactly returns the original mass N/Q .

This removes a spurious ℓ^1 loss from structured periodic reassembly. It does not remove an outer coefficient-mass loss and does not cover a nonperiodic physical multiplier, a generic additive phase, an arbitrary interval origin, deterministic prefixes or a four-point kernel. The conclusion is a good-scale L1 arithmetic interface, not positive L2, not an X -power saving and not a prime-pair statement.

1 The exact source contract

We use the general multiplicative-function form of Tao and Teräväinen [1, Theorem 3.1], not only its affine Liouville corollary. Let $X \geq 2$, let $g_1, g_2 : \mathbb{N} \rightarrow \mathbb{C}$ be one-bounded multiplicative functions, and let

$$1 \leq \mathcal{L} \leq \log X.$$

In the nonpretentious branch, $\delta_N = 0$ and

$$\exp M(g_1; X^2, \log^{1/125} X) \gg \mathcal{L}. \quad (1)$$

The equidistributed branch has a different hypothesis that will not be used in the quotient-Möbius application.

For a sufficiently small absolute $c > 0$, the source chooses a set

$$\mathcal{E}_X(g_1, g_2, \mathcal{L}) \subset [\sqrt{X}, X]$$

satisfying

$$\frac{1}{\log X} \int_{\mathcal{E}_X} \frac{dt}{t} \ll \mathcal{L}^{-c}. \quad (2)$$

For every $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X$, every

$$1 \leq W \leq \mathcal{L}^c, \quad b, h_1, h_2 = O(\mathcal{L}^c), \quad h_1 \neq h_2,$$

the conclusion is

$$\frac{W}{N} \left| \sum_{\substack{N < n \leq 2N \\ n \equiv b \pmod{W}}} (g_1(n + h_1) - \delta_N) g_2(n + h_2) \right| \ll \mathcal{L}^{-c}. \quad (3)$$

The exceptional set is selected before W, b, h_1, h_2 ; hence it is uniform in these parameters. The normalization W/N is part of the theorem.

Remark 1.1 (Local and global exceptional sets are different). Tao and Teräväinen [1, Remarks 3.2] also record a global Liouville affine special case, with one exceptional set for all sufficiently small polylogarithmic affine data. Equation (2) is the local set from the general theorem and may depend on $g_1, g_2, \mathcal{L}, X$. We never replace it by the global Liouville set.

2 An exact periodic residue partition

Fix a base modulus $Q \geq 1$, a residue $b_0 \pmod{Q}$, and a period $R \geq 1$. If $n \equiv b_0 \pmod{Q}$, write

$$z(n) = \frac{n - b_0}{Q} \in \mathbb{Z}.$$

Let $\rho : \mathbb{Z}/R\mathbb{Z} \rightarrow \mathbb{C}$.

Lemma 2.1 (Row-separated periodic partition). *For every finitely supported sequence $F(n)$,*

$$\sum_{n \equiv b_0 \pmod{Q}} F(n) \rho(z(n)) = \sum_{r \pmod{R}} \rho(r) \sum_{n \equiv b_0 + Qr \pmod{QR}} F(n). \quad (4)$$

The R refined classes are pairwise disjoint and their union is the base class $b_0 \pmod{Q}$.

Proof. The condition $z(n) \equiv r \pmod{R}$ is exactly $n \equiv b_0 + Qr \pmod{QR}$. Distinct $r \pmod{R}$ give distinct classes modulo QR , and every integer in the base class has one such r . $\square$

This is a pathwise partition, not a scalar replacement by a periodic density. It retains the value $\rho(r)$ on every refined class.

3 Periodic reassembly without a census loss

Theorem 3.1 (Source-native periodic reassembly). *Assume the hypotheses of (3) for fixed g_1, g_2 . Let $Q, R \geq 1$ satisfy*

$$QR \leq \mathcal{L}^c, \quad (5)$$

and suppose the residue representatives and shifts remain within the source's $O(\mathcal{L}^c)$ envelope. Then, for every $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X$,

$$\begin{aligned} \frac{Q}{N} \left| \sum_{\substack{N < n \leq 2N \\ n \equiv b_0 \pmod{Q}}} (g_1(n + h_1) - \delta_N) g_2(n + h_2) \right. \\ \left. \times \rho\left(\frac{n - b_0}{Q}\right) \right| \ll \|\rho\|_\infty \mathcal{L}^{-c}. \end{aligned} \quad (6)$$

The same exceptional set works for every allowed Q, R, b_0, ρ .

Proof. Apply lemma 2.1. Each inner sum is on one residue modulo QR . Choose its canonical representative in $\{0, \dots, QR - 1\}$; this remains inside the source height envelope because $QR \leq \mathcal{L}^c$. Equation (3), with $W = QR$, bounds its absolute value by

$$\frac{N}{QR} \mathcal{L}^{-c}.$$

There are R classes, and every coefficient has modulus at most $\|\rho\|_\infty$. Thus the unnormalized total is at most

$$R\|\rho\|_\infty \frac{N}{QR} \mathcal{L}^{-c} = \|\rho\|_\infty \frac{N}{Q} \mathcal{L}^{-c}.$$

Multiplication by Q/N proves (6). Uniformity of the source set in W, b, h_1, h_2 shows that no union of R exceptional sets is required. $\square$

Corollary 3.2 (Products of periodic masks). *Any finite product of bounded periodic masks may be combined into one bounded periodic function whose period divides the least common multiple of their periods. If the resulting QR satisfies (5), [theorem 3.1](#) charges its supremum norm but no separate residue-count exponent.*

This applies, for example, to an exactly periodic coprimality mask or a fixed-period coefficient. A rational additive phase is included only when its exact denominator is absorbed in R . No conclusion is made for a generic real phase.

4 What is and is not an ℓ^1 cost

The cancellation in [theorem 3.1](#) is a normalization fact:

$$R \times \frac{N}{QR} = \frac{N}{Q}. \quad (7)$$

It does not say that every finite decomposition has no cost. If several distinct outer records carry coefficients γ_ξ , then a triangle-inequality return still involves the actual ratio

$$\frac{\sum_\xi |\gamma_\xi| (N_\xi/Q_\xi)}{\text{declared physical comparison mass}}.$$

That ratio must be measured on the literal archive. Likewise, splitting a nonperiodic multiplier into unrelated pieces does not satisfy (7).

Proposition 4.1 (No prefix promotion). *Equations (2)–(6) do not imply a bound for a prescribed deterministic scale, an arbitrary interval $(A, B]$, or the maximum over all initial prefixes of such an interval.*

Proof. The source conclusion is asserted only for $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X$ and for the interval $(N, 2N]$. A prescribed scale may lie in $\mathcal{E}_X$, while an arbitrary interval or prefix is not among the theorem’s quantified objects. None is created by the exact residue identity. $\square$

5 Certificate and claim boundary

The deterministic certificate checks the row-separated partition, the native W/N normalization and the exact mass identity (7). It also rejects mutations that replace the local exceptional set by the global Liouville set, move QR outside the source envelope, or attach unsupported weights and prefixes. These are finite consistency checks, not numerical evidence for (3).

Statement	Level and status
Periodic residue partition	Exact L0, PROVED.
No R -census loss under $QR \leq \mathcal{L}^c$	Sourced good-scale L1, PROVED.
One exceptional set for all residues and periods	PROVED for fixed $g_1, g_2, X, \mathcal{L}$.
Nonperiodic physical multiplier or generic phase	OPEN; not in the theorem.
Arbitrary origin or all-prefix maximum	OPEN; no implication.
Positive L2, fixed X -power or $1/400$	Not proved.
Prime-pair or twin-prime theorem	Not proved.

TPC-147 therefore removes one false reassembly loss while preserving all source quantifiers. The next paper constructs multiplicative functions whose values on a determinant-two progression are the literal Möbius quotients.

References

- [1] Terence Tao and Joni Teräväinen. Quantitative Correlations and Some Problems on Prime Factors of Consecutive Integers, 2026.